Toric Certificates for Effective Skolem Tails and Subspace Orbit Reachability

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Abstract

We develop a certificate framework for proving effective zero-free tails for scalar and vector algebraic linear recurrence sequences. The framework applies to the Skolem Problem, the Simultaneous Skolem Problem, and certified instances of the Subspace Orbit Problem.

After passing to a common Binet representation, splitting root-of-unity degeneracies, saturating residual torsion, and peeling globally cancelled dominant shells, each branch of a vector recurrence is put in the form

$$\mathbf{u}_{Mk+r} = \Lambda^k \big(\mathbf{D}(k, z(k)) + \mathbf{E}(k)\big),$$

where

$$z(k) = c\eta^k \in \mathbb{T}^\rho, \qquad \mathbf{D}(T, X) \in L[T][X_1^{\pm 1}, \dots, X_\rho^{\pm 1}]^m,$$

and

$$|\mathbf{E}(k)| \le C_E (k+1)^s \theta^k, \qquad 0 < \theta < 1.$$

The toric rank, the order of the recurrence, the number of dominant roots, and the root multiplicities are unrestricted.

The main theorem is a soundness theorem. If the dominant vector admits a verified toric lower-bound certificate, then all zeros of the branch lie below a computable bound unless the branch is identically zero. The remaining prefix is checked exactly. We give several verifiable certificate formats: joint semialgebraic certificates, Positivstellensatz and sum-of-squares certificates, Bezout–toric certificates, and scalar character-polynomial certificates. We then describe a partial compiler, Factor–Reduce–Certify, which constructs certificates automatically for leading-vector zero-free branches and for quotient-character factors.

The method is sound but incomplete: failure to certify an instance does not imply anything about its zero set. The contribution is not a solution of the general Skolem Problem. It is a checkable and partially automatic mechanism for proving effective Skolem tails in high-rank, high-order families, including vector examples where scalar coordinatewise certification fails.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, ChatGPT, by OpenAI, was used to assist with mathematical drafting, formalization, review, and editing. This work is shared as a preliminary AI-assisted mathematical note. The mathematical content may have been only partially reviewed and may contain errors; it should not be treated as peer-reviewed or as a fully verified manuscript.

1 Introduction

Let K be a number field. A scalar linear recurrence sequence over K is a sequence $u = (u_n)_{n\geq 0}$ satisfying

$$u_{n+d} = a_{d-1}u_{n+d-1} + \cdots + a_0 u_n$$

for some $a_0, \ldots, a_{d-1} \in K$. The Skolem Problem asks whether there exists $n \geq 0$ such that

$$u_n = 0.$$

The strong computational form asks to compute the entire zero set

$$Z(u) = \{n \geq 0 : u_n = 0\}.$$

The Skolem–Mahler–Lech theorem asserts that $Z(u)$ is a finite union of arithmetic progressions together with a finite exceptional set. Its known proofs do not provide a general algorithm for computing the finite exceptional set. The computational heart of Skolem is therefore the construction of an effective zero-free tail.

This paper studies such tails through toric certificates. The scalar case is included, but the natural level of generality is vectorial. Let

$$\mathbf{u}_n = (u_1(n), \ldots, u_m(n)) \in K^m$$

be a vector of algebraic linear recurrence sequences. The simultaneous zero set is

$$Z(\mathbf{u}) = \{n \geq 0 : \mathbf{u}_n = 0\}.$$

This formulation covers the Simultaneous Skolem Problem. It also covers affine subspace orbit reachability: if $A \in K^{d\times d}$, $x_0 \in K^d$, and

$$V = \{y \in K^d : By = b\},$$

then

$$A^n x_0 \in V \iff BA^n x_0 - b = 0.$$

The central idea is to separate the dominant vector from zero. On each branch, after standard spectral reductions, we obtain

$$\mathbf{u}_{Mk+r} = \Lambda^k(\mathbf{D}(k, z(k)) + \mathbf{E}(k)),$$

where $z(k)$ is a toric orbit on a compact toric coset, $\mathbf{D}$ is the normalized dominant vector, and $\mathbf{E}$ is exponentially smaller. If one can certify that

$$\mathbf{D}(k, z(k)) \neq 0 \implies |\mathbf{D}(k, z(k))| \geq \exp(-C(\log(k+2))^A)$$

outside a computable prefix, then the exponentially small error cannot cancel the dominant vector outside that prefix. The rest is an exact finite computation.

The vector viewpoint is essential. A scalar coordinate may contain a factor such as

$$X + Y - 1,$$

which is outside the automatic one-character Baker class. But a system such as

$$(X + Y - 1,\ X - Y)$$

has no common zero on $\mathbb{T}^2$, and the norm of the vector is bounded away from zero by a semialgebraic certificate. Thus the vector method proves effective tails in cases not visible to scalar coordinatewise methods.

The contributions are:

1. a global vector dominant-shell normal form with torsion saturation;
2. a verifier theorem for toric lower-bound certificates;
3. joint semialgebraic and Positivstellensatz/SOS certificates;
4. Bezout–toric certificates converting scalar lower bounds into vector lower bounds;
5. a scalar Baker–Matveev lemma for expressions $F(k, \delta^k)$;
6. quotient-character reduction, producing scalar atoms automatically from Laurent factors whose exponent supports become one-dimensional modulo the toric relation lattice;
7. a partial Factor–Reduce–Certify compiler;
8. applications to Simultaneous Skolem, Subspace Orbit, affine linear-loop reachability, algebraic target hitting, and eventual cone membership.

The framework is intentionally conservative. It proves effective tails for certified toric dominant shells. It does not decide arbitrary toric shells of rank at least two and does not solve the general Skolem Problem.

2 Effective input and certificate model

An input consists of:

1. a number field K, represented effectively;
2. either a scalar or vector linear recurrence sequence over K, or a linear dynamical system (A, x_0, B, b);
3. a complex embedding $\sigma : K \hookrightarrow \mathbb{C}$.

All computations take place in finite extensions of K. We use standard exact algorithms in number fields: arithmetic, factorization, construction of splitting fields, root-of-unity testing, exact zero testing, comparison of algebraic absolute values under σ, and computation of multiplicative relation lattices. These algorithms are not claimed to be efficient in general; they are used as exact symbolic subroutines.

A certificate is extended input. The algorithms in this paper are sound: if the certificate verifies, then the output zero set is correct. They are not complete: failure to certify means "not certified", not "no zero" and not "undecidable".

We separate three layers.

- The verifier theorem: a valid certificate implies an effective zero-free tail.
- Automatic certificate generation: for certain structural classes, the certificate is constructed by the compiler.
- Positive search: for some certificates, such as Bezout–toric identities of bounded degree, one may search positively, but failure of the search has no mathematical consequence.

3 Vector Binet form and global shells

Let

$$\mathbf{u}_n = (u_1(n), \dots, u_m(n)) \in K^m$$

be a vector of algebraic linear recurrence sequences. Passing to a common splitting field L/K, we may write

$$\mathbf{u}_n = \sum_{\lambda \in \mathcal{R}} \mathbf{P}_\lambda(n)\lambda^n,$$

where $\mathcal{R} \subset L^\times$ is finite and

$$\mathbf{P}_\lambda(T) \in L[T]^m.$$

Zero characteristic roots contribute only to a finite initial segment. That segment is checked exactly, and the sequence is shifted. Hence all roots in the remaining Binet representation are nonzero.

Fix an embedding

$$\sigma : L \hookrightarrow \mathbb{C}.$$

A global vector shell is a set

$$S_\rho = \{\lambda \in \mathcal{R} : |\sigma(\lambda)| = \rho\}.$$

The shell contribution is

$$\mathbf{U}_\rho(n) = \sum_{\lambda \in S_\rho} \mathbf{P}_\lambda(n)\lambda^n.$$

The shell is vectorially cancelled if

$$\mathbf{U}_\rho(n) \equiv 0$$

as a vector-valued exponential-polynomial function. This means every coordinate is identically zero.

After torsion splitting, distinct bases with no root-of-unity quotient give linearly independent exponential-polynomial functions. Therefore vectorial cancellation is decidable exactly.

4 Torsion splitting and torsion saturation

If λ/μ is a root of unity, let $o(\lambda, \mu)$ be its order. Let

$$M_0 = \operatorname{lcm}\{o(\lambda, \mu) : \lambda, \mu \in \mathcal{R},\ \lambda/\mu \in \mu_\infty\}.$$

For each residue class $r \pmod{M_0}$, define

$$\mathbf{v}_k^{(r)} = \mathbf{u}_{M_0 k + r}.$$

Equal bases in the resulting Binet representation are combined exactly.

A second torsion issue remains. A shell with relative multiplicative rank ρ modulo torsion need not be an exact toric coset. Its roots have the form

$$\lambda = \Lambda\, \xi_\lambda\, \gamma^a,$$

where ξ_λ is a root of unity and

$$\gamma^a = \gamma_1^{a_1} \cdots \gamma_\rho^{a_\rho}, \qquad a \in \mathbb{Z}^\rho.$$

Let

$$Q = \mathrm{lcm}_\lambda \,\mathrm{ord}(\xi_\lambda).$$

Passing to subbranches

$$k = Qh + s$$

makes

$$\xi_\lambda^k = \xi_\lambda^s$$

constant. The polynomial coefficients are transformed by

$$\mathbf{P}_\lambda(k) \mapsto \mathbf{P}_\lambda(Qh + s),$$

and the bases by

$$\lambda^k \mapsto \lambda^s(\lambda^Q)^h.$$

All new data are computable. We call this step torsion saturation.

5 Vector toric normal form

After torsion splitting and torsion saturation, peel globally cancelled shells until the first non-cancelled global shell is reached. Let S be that shell. Choose $\Lambda \in S$. The roots in S can be written as

$$\lambda = \Lambda\gamma^a, \qquad a \in A \subset \mathbb{Z}^\rho.$$

Since all roots in S have the same σ-absolute value, the toric parameters satisfy

$$|\sigma(\gamma_j)| = 1.$$

Define

$$z(k) = (\gamma_1^k, \ldots, \gamma_\rho^k).$$

Then the contribution of S has the form

$$\Lambda^k \mathbf{D}(k, z(k)),$$

where

$$\mathbf{D}(T, X) \in L[T][X_1^{\pm 1}, \ldots, X_\rho^{\pm 1}]^m.$$

All lower shells have strictly smaller σ-absolute value. Hence the branch has normal form

$$\mathbf{v}_k = \Lambda^k\big(\mathbf{D}(k, z(k)) + \mathbf{E}(k)\big),$$

with

$$|\mathbf{E}(k)| \le C_E(k+1)^s\theta^k, \qquad 0 < \theta < 1.$$

Here

$$|\mathbf{w}|^2 = \sum_{i=1}^m |\sigma(w_i)|^2.$$

6 Toric cosets

After reindexing, it is convenient to write

$$z(k) = c\eta^k = (c_1\eta_1^k, \dots, c_\rho\eta_\rho^k),$$

with

$$|\sigma(c_j)| = |\sigma(\eta_j)| = 1.$$

Define the exact relation lattice

$$\mathcal{L}_\eta = \{a = (a_1, \dots, a_\rho) \in \mathbb{Z}^\rho : \eta^a = 1\},$$

where

$$\eta^a = \eta_1^{a_1} \cdots \eta_\rho^{a_\rho}.$$

This lattice is computable.

Let

$$T_\eta = \{y \in \mathbb{T}^\rho : y^a = 1 \text{ for all } a \in \mathcal{L}_\eta\}.$$

The orbit closure of $z(k) = c\eta^k$ is the compact toric coset

$$C = cT_\eta.$$

Equivalently,

$$C = \{x \in \mathbb{T}^\rho : x^a = c^a \text{ for all } a \in \mathcal{L}_\eta\}.$$

We do not assume T_η is connected. If finite torsion remains in η, T_η may have finitely many components. The equations above describe the full compact coset.

For effective real-algebraic verification, write

$$X_j = x_j + iy_j.$$

The torus is defined by

$$x_j^2 + y_j^2 = 1,$$

and the coset equations are the real and imaginary parts of

$$X^a = c^a$$

for a basis of $\mathcal{L}_\eta$.

7 The certified-tail theorem

Theorem 7.1 (Certified-tail soundness)**.** *Let a saturated branch have vector toric normal form*

$$\mathbf{v}_k = \Lambda^k\big(\mathbf{D}(k, z(k)) + \mathbf{E}(k)\big),$$

where

$$|\mathbf{E}(k)| \le C_E(k+1)^s\theta^k, \qquad 0 < \theta < 1.$$

Assume there are computable constants A, C, N_0 such that, for every $k \ge N_0$,

$$\mathbf{D}(k, z(k)) \ne 0 \quad \Longrightarrow \quad |\mathbf{D}(k, z(k))| \ge \exp(-C(\log(k+2))^A).$$

Assume also that exact zeros of

$$\mathbf{D}(k, z(k))$$

are either contained in a computable prefix or occur only on subbranches where the global dominant vector is identically cancelled and hence peeled.

Then the branch has a computable zero-free tail. Consequently the zeros on the branch are computable exactly as a finite set plus any identically zero subprogressions.

Proof. Choose $B \geq N_0$ such that

$$C_E(k+1)^s \theta^k < \frac{1}{2} \exp(-C(\log(k+2))^A)$$

for all $k \geq B$. Such B is computable. For $k \geq B$, if $\mathbf{D}(k, z(k)) \neq 0$, then

$$|\mathbf{D}(k, z(k)) + \mathbf{E}(k)| \geq |\mathbf{D}(k, z(k))| - |\mathbf{E}(k)| > 0.$$

Hence $\mathbf{v}_k \neq 0$. The remaining possible zeros are in a computable prefix or in identically zero subbranches produced by peeling. The finite prefix is checked exactly in L. □

8 Joint semialgebraic certificates

Definition 8.1 (Joint semialgebraic certificate)**.** *Let*

$$\mathbf{D}(T, X) = (D_1(T, X), \dots, D_m(T, X)).$$

A joint semialgebraic certificate, or joint S-certificate, on a toric coset $C = cT_\eta$ consists of explicit data

$$B_0 \in \mathbb{N}, \qquad L_0 \in \mathbb{N}, \qquad q \in \mathbb{Q}_{>0}$$

such that

$$\sum_{i=1}^{m} |\sigma(D_i(t, x))|^2 \geq q(t+1)^{-L_0}$$

for every

$$t \geq B_0, \qquad x \in C.$$

The assertion is a universal formula over a real closed field. Laurent monomials are represented on the torus by

$$X_j = x_j + iy_j, \qquad X_j^{-1} = x_j - iy_j.$$

The coset C is represented by polynomial equations

$$x_j^2 + y_j^2 = 1$$

and the real and imaginary parts of

$$X^a = c^a$$

for a basis of $\mathcal{L}_\eta$.

The certificate condition becomes

$$\forall t, x, y : \ (t \geq B_0 \wedge (x + iy) \in C) \Rightarrow (t+1)^{L_0} \sum_i |\sigma(D_i(t, x + iy))|^2 - q \geq 0.$$

This formula is effectively decidable by quantifier elimination over real closed fields.

Proposition 8.2. *A joint S-certificate implies the hypothesis of the certified-tail theorem with a polynomial lower bound:*

$$|\mathbf{D}(k, z(k))| \geq q^{1/2}(k+1)^{-L_0/2}.$$

Proof. Since $z(k) \in C$, substituting $t = k$ and $x = z(k)$ into the certificate gives the result. □

9 SOS and Positivstellensatz certificates

Quantifier elimination proves decidability, but one may also use algebraic certificates that are easier to verify.

Let C be represented by equalities

$$f_1 = \cdots = f_r = 0$$

and inequalities

$$g_1 \geq 0, \ldots, g_s \geq 0.$$

These include the torus equations, the coset equations, and the tail condition $t - B_0 \geq 0$. A Positivstellensatz/SOS certificate for the joint lower bound consists of an identity

$$(t+1)^L \sum_i |D_i(t,X)|^2 - q = \sigma_0 + \sum_j \sigma_j g_j + \sum_\ell h_\ell f_\ell,$$

where the σ_j are sums of squares and the h_ℓ are arbitrary polynomials, all with algebraic coefficients.

Such an identity is checked by exact polynomial arithmetic and by verifying the supplied sum-of-squares decompositions. When present, it gives a joint S-certificate.

This SOS formulation is not required for decidability, but it gives a more explicit certificate format than quantifier elimination.

10 Automatic joint certificates from leading vectors

Write

$$\mathbf{D}(T,X) = T^d \mathbf{D}_d(X) + T^{d-1}\mathbf{D}_{d-1}(X) + \cdots + \mathbf{D}_0(X),$$

with $\mathbf{D}_d \neq 0$.

Proposition 10.1 (Leading-vector criterion)**.** *If*

$$\mathbf{D}_d(x) \neq 0 \qquad \textit{for every } x \in C,$$

then a joint S-certificate exists and can be effectively found.

Proof. The function

$$x \mapsto |\mathbf{D}_d(x)|^2$$

is continuous and positive on the compact coset C. Its positivity can be verified by quantifier elimination. Once verified, one can find a rational $q_0 > 0$ such that

$$|\mathbf{D}_d(x)|^2 \geq q_0$$

for all $x \in C$.

For $j < d$, compute an upper bound M_j for $|\mathbf{D}_j(x)|$ on C. Since monomials have modulus 1 on the torus, summing absolute values of algebraic coefficients gives an effective upper bound.

For t large enough,

$$\left| \sum_{j<d} t^j \mathbf{D}_j(x) \right| \leq \frac{1}{2} q_0^{1/2} t^d$$

uniformly on C. Hence

$$|\mathbf{D}(t,x)| \geq \frac{1}{2} q_0^{1/2} t^d.$$

This is a joint S-certificate. $\square$

11 Scalar character-polynomial atoms

A scalar atom is an expression

$$F(T, \chi(X)),$$

where

$$F(T, Y) \in L[T, Y] \setminus \{0\}$$

and

$$\chi(X) = X_1^{a_1} \cdots X_\rho^{a_\rho}$$

is a toric character. Along the branch

$$z(k) = c\eta^k,$$

we have

$$\chi(z(k)) = \chi(c)\chi(\eta)^k.$$

Absorbing the constant $\chi(c)$ into F, one reduces to

$$F(k, \delta^k), \qquad \delta = \chi(\eta).$$

12 The Baker–Matveev lemma

We use the following standard consequence of effective lower bounds for linear forms in logarithms.

Lemma 12.1 (Scalar character lower bound)**.** *Let*

$$F(T, Y) \in L[T, Y] \setminus \{0\},$$

let

$$\delta \in L^\times, \qquad |\sigma(\delta)| = 1,$$

and suppose δ is not a root of unity. Then there exist computable constants

$$C > 0, \qquad N_0 \in \mathbb{N}$$

such that, for every $k \geq N_0$,

$$F(k, \delta^k) \neq 0 \quad \Longrightarrow \quad |\sigma(F(k, \delta^k))| \geq \exp(-C(\log(k+2))^2).$$

Moreover, exact zeros of $F(k, \delta^k)$ are contained in a computable finite prefix.

Proof. Write

$$F(T, Y) = \sum_{j=0}^{r} f_j(T) Y^j.$$

For each k, let

$$F_k(Y) = F(k, Y).$$

The set of k such that $F_k \equiv 0$ is finite and computable unless all f_j vanish identically, which is excluded. The case in which F_k is constant nonzero is harmless and gives a Liouville lower bound.

Assume F_k is nonconstant. Factor

$$F_k(Y) = a_k \prod_\nu (Y - \eta_{\nu,k})$$

over $\overline{\mathbb{Q}}$. The degree of F_k is at most r, though it may drop for special k. This only decreases the number of factors.

The coefficients $f_j(k)$ have logarithmic height $O(\log(k+2))$. Standard height estimates for roots of a polynomial give

$$h(\eta_{\nu,k}) \le C_1 \log(k+2)$$

with C_1 effective. Liouville's inequality gives

$$|\sigma(a_k)| \ge \exp(-C_2 \log(k+2))$$

whenever $a_k \ne 0$.

Let $\eta = \eta_{\nu,k}$. If $\eta = 0$, then

$$|\sigma(\delta^k - \eta)| = 1.$$

Assume $\eta \ne 0$ and $\delta^k \ne \eta$. Apply an effective lower bound for linear forms in logarithms to

$$\eta^{-1}\delta^k - 1.$$

The degree of the field $\mathbb{Q}(\delta, \eta)$ is uniformly bounded in terms of L and r. The height of η is $O(\log k)$, and the exponent of δ is k. A standard Baker–Wuestholz or Matveev estimate yields

$$|\sigma(\eta^{-1}\delta^k) - 1| \ge \exp(-C_3(\log(k+2))^2).$$

Liouville gives

$$|\sigma(\eta)| \ge \exp(-C_4 \log(k+2)).$$

Therefore

$$|\sigma(\delta^k - \eta)| \ge \exp(-C_5(\log(k+2))^2).$$

If $F(k, \delta^k) \ne 0$, no factor vanishes. Multiplying the lower bounds for the leading coefficient and all nonzero factors gives

$$|\sigma(F(k, \delta^k))| \ge \exp(-C(\log(k+2))^2).$$

Now suppose

$$F(k, \delta^k) = 0.$$

Then δ^k is a root of F_k, so

$$h(\delta^k) \le C_1 \log(k+2).$$

Since δ is not a root of unity, $h(\delta) > 0$, and

$$h(\delta^k) = kh(\delta).$$

Thus

$$kh(\delta) \le C_1 \log(k+2),$$

which fails beyond a computable bound. $\square$

13 Torsion scalar atoms

If δ is a root of unity of order Q, pass to subbranches

$$k = Qh + s.$$

Then

$$F(k, \delta^k) = F(Qh + s, \delta^s) = G_s(h),$$

where $G_s \in L[h]$.

If $G_s \not\equiv 0$, its integer zeros are finite and computable, and Liouville gives a polynomial lower bound for nonzero values.

If $G_s \equiv 0$, the scalar atom vanishes identically on that subbranch. In a scalar product certificate this may force the dominant scalar to vanish. In a vector or Bezout certificate it means only that the particular scalar test has failed. The algorithm either peels the branch if the full dominant vector is identically zero, or requires a new certificate on that subbranch.

Termination follows because every peeling removes one global shell and all splitting operations are finite.

14 Bezout–toric certificates

Let

$$\mathbf{D} = (D_1, \dots, D_m).$$

Let

$$a^{(1)}, \dots, a^{(\ell)}$$

be a basis of $\mathcal{L}_\eta$, and define the coset binomials

$$G_j(X) = X^{a^{(j)}} - c^{a^{(j)}}.$$

Then

$$G_j(z(k)) = 0.$$

Definition 14.1 (Bezout–toric certificate)**.** *A Bezout–toric certificate consists of Laurent polynomials*

$$A_i(T, X), \quad B_j(T, X), \quad M(T, X)$$

such that

$$M(T, X) = \sum_{i=1}^{m} A_i(T, X) D_i(T, X) + \sum_{j=1}^{\ell} B_j(T, X) G_j(X),$$

together with a scalar certificate for $M(k, z(k))$ giving

$$|M(k, z(k))| \geq \exp(-C(\log(k+2))^A)$$

outside a computable prefix.

The identity is verified exactly in the Laurent polynomial ring.

Lemma 14.2. *There exist computable constants C_A, R such that*

$$\left(\sum_i |A_i(k, z(k))|^2 \right)^{1/2} \leq C_A (k+1)^R.$$

Proof. Write
$$A_i(T,X)=\sum_{\nu} a_{i,\nu}(T)X^{b_{i,\nu}}.$$
On the torus,
$$|X^{b_{i,\nu}}|=1.$$
Each polynomial $a_{i,\nu}(k)$ grows at most polynomially in k. Summing finitely many terms gives the stated bound. $\square$

Proposition 14.3. *A Bezout–toric certificate implies the lower-bound hypothesis of the certified-tail theorem.*

Proof. Along the orbit,
$$M(k,z(k))=\sum_i A_i(k,z(k))D_i(k,z(k)).$$
By Cauchy–Schwarz,
$$|M(k,z(k))|\le\left(\sum_i |A_i(k,z(k))|^2\right)^{1/2}|\mathbf{D}(k,z(k))|.$$
Using the preceding lemma,
$$|M(k,z(k))|\le C_A(k+1)^R|\mathbf{D}(k,z(k))|.$$
A subexponential lower bound for M therefore gives a subexponential lower bound for $|\mathbf{D}|$. $\square$

One may search for Bezout–toric certificates by fixing degree bounds for the unknown A_i, B_j, M, imposing the identity as a finite linear system over L, and checking whether M admits a scalar S/B/T certificate. If the search succeeds, the certificate is valid. If it fails at a chosen bound, no conclusion follows. Thus this is a positive semidecision procedure, not a complete algorithm.

15 Quotient-character reduction

Let
$$H(T,X)=\sum_{a\in A} h_a(T)X^a$$
be a Laurent factor. Consider the image of A in
$$\mathbb{Z}^\rho/\mathcal{L}_\eta.$$
We say that H has quotient affine rank at most one if the differences
$$a-a_0,\qquad a\in A,$$
span a vector space of dimension at most one in
$$(\mathbb{Z}^\rho/\mathcal{L}_\eta)\otimes_{\mathbb{Z}}\mathbb{Q}.$$

Proposition 15.1. *If H has quotient affine rank at most one, then after finite torsion splitting,*
$$H(k,z(k))=\alpha_0^kF(k,\delta^k)$$
for some algebraic α_0,δ of σ-absolute value 1 and some
$$F(T,Y)\in L[T,Y].$$

Proof. Choose $a_0 \in A$. Since the quotient affine rank is at most one, there exists $v \in \mathbb{Z}^\rho$ such that every $a - a_0$ maps to a rational multiple of v in the quotient. Clearing denominators and splitting finite torsion in the quotient, we may assume that for each $a \in A$,

$$a = a_0 + r_a v + \ell_a$$

with $r_a \in \mathbb{Z}$ and $\ell_a \in \mathcal{L}_\eta$.

On the coset $z(k) = c\eta^k$,

$$z(k)^a = c^a \eta^{ak} = c^a \eta^{a_0 k} (\eta^v)^{r_a k},$$

because $\eta^{\ell_a} = 1$. Thus

$$H(k, z(k)) = \eta^{a_0 k} \sum_{a \in A} h_a(k) c^a (\eta^v)^{r_a k}.$$

After multiplying by a power of Y if necessary, the sum becomes $F(k, \delta^k)$ with

$$\delta = \eta^v.$$

The factor $\eta^{a_0 k}$ is α_0^k and has modulus 1. $\square$

Corollary 15.2. *A quotient-affine-rank-one factor is automatically converted into a scalar Baker or torsion atom.*

16 The Factor–Reduce–Certify compiler

The partial compiler operates branchwise.

1. Compute the vector toric normal form.
2. Compute the toric coset $C = cT_\eta$.
3. Attempt the leading-vector criterion.
4. If it fails, attempt to construct a Bezout–toric certificate by bounded-degree linear algebra.
5. For scalar factors or scalar tests, factor in the Laurent polynomial ring.
6. For each irreducible factor, compute its quotient affine rank modulo $\mathcal{L}_\eta$.
7. If the rank is at most one, reduce it to a scalar atom.
8. Verify remaining S-certificates by quantifier elimination, SOS identities, or Positivstellensatz identities.
9. If every branch is certified, compute the zero-free tails and exact prefixes.
10. If some branch is not certified, return “not certified”.

Theorem 16.1 (Compiler soundness)**.** *If Factor–Reduce–Certify succeeds on all branches of a vector recurrence* $\mathbf{u}$*, then it computes*

$$Z(\mathbf{u}) = \{n \geq 0 : \mathbf{u}_n = 0\}$$

exactly as a finite union of arithmetic progressions and a finite set.

Proof. Every successful branch satisfies the certified-tail theorem through a joint S-certificate, a Bezout–toric certificate, or scalar atoms combined into a vector certificate. Identically zero branches give progressions. Nonzero certified branches have zero-free tails and exact finite prefix checks. $\square$

17 Applications

Theorem 17.1 (Simultaneous Skolem). *Let*

$$u^{(1)}, \dots, u^{(m)}$$

be algebraic linear recurrence sequences and set

$$\mathbf{u}_n = (u_n^{(1)}, \dots, u_n^{(m)}).$$

If Factor–Reduce–Certify succeeds on $\mathbf{u}$, *then*

$$\{n \geq 0 : u_n^{(1)} = \cdots = u_n^{(m)} = 0\}$$

is computable exactly.

This is stronger than separately solving the scalar Skolem instances and intersecting their zero sets, because a joint vector certificate may exist even when a scalar coordinate is outside the automatic scalar certificate class.

Theorem 17.2 (Subspace Orbit and affine linear loops). *Let*

$$A \in K^{d \times d}, \qquad x_0 \in K^d,$$

and let

$$V = \{y \in K^d : By = b\}$$

be an affine subspace. Define

$$\mathbf{u}_n = BA^n x_0 - b.$$

If Factor–Reduce–Certify succeeds on $\mathbf{u}_n = BA^n x_0 - b$, *then the hitting-time set*

$$\{n \geq 0 : A^n x_0 \in V\}$$

is computable exactly.

Proof. We have

$$A^n x_0 \in V \iff BA^n x_0 - b = 0.$$

Apply compiler soundness. □

This gives a certified reachability theorem for affine linear loops

$$x := Ax$$

with target condition $Bx = b$.

Corollary 17.3 (Algebraic target hitting). *Let*

$$W = \{y : P_1(y) = \cdots = P_m(y) = 0\}$$

be an affine algebraic set over K. *Since coordinates of* $A^n x_0$ *are linear recurrence sequences and such sequences are closed under addition and pointwise multiplication, each sequence*

$$P_i(A^n x_0)$$

is a linear recurrence sequence. If Factor–Reduce–Certify succeeds on

$$(P_1(A^n x_0), \dots, P_m(A^n x_0)),$$

then

$$\{n \geq 0 : A^n x_0 \in W\}$$

is computable exactly.

18 Eventual cone membership and ultimate positivity

Let $C \subseteq \mathbb{R}^m$ be a semialgebraic cone or a closed semialgebraic set. Suppose a real vector recurrence has normal form

$$\mathbf{u}_k = \Lambda^k(\mathbf{D}(k, z(k)) + \mathbf{E}(k))$$

with $\Lambda > 0$, and suppose a certificate proves that $\mathbf{D}(k, z(k))$ has distance at least

$$c(k+1)^{-L}$$

from the boundary of C for $k \geq B$, with a fixed side of the boundary. Then $\mathbf{u}_k \in C$ eventually if and only if the dominant vector lies on that side. The finite prefix is checked exactly.

For $m = 1$ and $C = [0, \infty)$, this gives certified ultimate positivity.

19 A high-rank vector family

Let $\rho \geq 2$, and let

$$\gamma_1, \ldots, \gamma_\rho \in L^\times$$

satisfy

$$|\sigma(\gamma_i)| = 1$$

and be multiplicatively independent modulo torsion.

Define

$$\mathbf{D}_\rho(X_1, \ldots, X_\rho) = (X_2 - X_1,\, X_3 - X_1,\, \ldots,\, X_\rho - X_1,\, X_1 + \cdots + X_\rho - 1)\,.$$

Proposition 19.1. *The vector $\mathbf{D}_\rho$ has no zero on $\mathbb{T}^\rho$.*

Proof. If $\mathbf{D}_\rho(X) = 0$, then

$$X_2 = X_1, \quad \ldots, \quad X_\rho = X_1.$$

Thus

$$X_1 = \cdots = X_\rho.$$

The last component gives

$$\rho X_1 = 1.$$

But $|X_1| = 1$, whereas $|1/\rho| < 1$ for $\rho \geq 2$. This is impossible. □

Therefore

$$Q_\rho(X) = |\mathbf{D}_\rho(X)|^2$$

has a strictly positive minimum on $\mathbb{T}^\rho$, and a joint S-certificate can be verified.

Let

$$\mathbf{u}_n = \Lambda^n \left(\mathbf{D}_\rho(\gamma_1^n, \ldots, \gamma_\rho^n) + \mathbf{E}(n)\right),$$

where

$$|\mathbf{E}(n)| \leq C\theta^n, \qquad 0 < \theta < 1.$$

Then Factor–Reduce–Certify succeeds and computes the simultaneous zero set.

The scalar coordinate

$$X_1 + \cdots + X_\rho - 1$$

has exponent support of affine rank ρ, so it is not an automatic quotient-character scalar factor. The vector system is nevertheless certified by joint separation. This demonstrates the genuine advantage of the vector method.

20 Realization as Subspace Orbit

Let

$$A = \operatorname{diag}(\gamma_1, \ldots, \gamma_\rho), \qquad x_0 = (1, \ldots, 1).$$

Then

$$A^n x_0 = (\gamma_1^n, \ldots, \gamma_\rho^n).$$

Let $V_\rho \subseteq K^\rho$ be the affine subspace defined by

$$x_2 - x_1 = 0, \quad x_3 - x_1 = 0, \quad \ldots, \quad x_\rho - x_1 = 0,$$

and

$$x_1 + \cdots + x_\rho = 1.$$

Then

$$A^n x_0 \in V_\rho \quad \Longleftrightarrow \quad \mathbf{D}_\rho(\gamma_1^n, \ldots, \gamma_\rho^n) = 0.$$

By the preceding proposition, the pure dominant model has no hitting time. With exponentially smaller subdominant perturbations, the subspace-orbit certificate theorem gives a computable zero-free tail and exact prefix check.

This gives a high-rank family of certified Subspace Orbit instances in which one scalar equation contains high-rank toric support, but the vector system is jointly separated.

21 Relation to previous work

The Skolem Problem is open in general. The Skolem–Mahler–Lech theorem gives the eventual structure of the zero set but is not a general algorithm for computing the finite exceptional set.

Low-order and few-dominant-root results decide important special cases. The present framework is different: it is not based on low order or few dominant roots, but on a verifiable toric separation certificate.

Modern algorithms for broader recurrence classes often produce correct certificates when they terminate, with termination proved under strong Diophantine conjectures. The present method is unconditional inside the certified class.

The Evertse–Schlickewei–Schmidt theory gives explicit upper bounds for the number of non-degenerate solutions of linear equations in variables from a finite-rank multiplicative group [10]. Such results are close in spirit, but a bound on the number of solutions does not by itself give a computable bound on the positions of the indices n. The present method produces an explicit zero-free tail once a certificate is verified.

Bacik and Varonka study Subspace Orbit and Simultaneous Skolem from a dimension-theoretic perspective [5]. Their results are orthogonal to the present work. They prove decidability in certain target-dimension regimes and hardness in others. The present framework imposes no a priori target-dimension threshold; it requires a toric separation certificate for the dominant vector.

The vector feature is essential. The family above shows that scalar coordinatewise certification can fail while joint vector certification succeeds.

22 Limitations

The framework does not decide arbitrary toric shells.

A scalar expression such as

$$X + Y - 1$$

on a genuinely rank-two orbit is not automatically handled by the scalar quotient-character method. It can be handled only if it appears in a joint vector system with a joint S-certificate, a Bezout–toric certificate, or some other verified lower-bound mechanism.

Thus the result is

effective tails for certified toric dominant shells,

not

decidability for all toric dominant shells.

The compiler is sound but incomplete.

23 Conclusion

We have developed a certificate framework for effective zero-free tails in scalar and vector linear recurrence sequences. The main novelty is the vector toric viewpoint: for Simultaneous Skolem and Subspace Orbit, it is enough to separate the dominant vector from zero. This can succeed even when scalar coordinatewise methods fail.

The framework includes:

1. global vector dominant-shell normal form;
2. torsion saturation and toric coset closures;
3. certified-tail soundness;
4. joint semialgebraic and SOS certificates;
5. Bezout–toric certificates;
6. scalar Baker character-polynomial atoms;
7. quotient-character reduction;
8. a partial Factor–Reduce–Certify compiler;
9. applications to Simultaneous Skolem, Subspace Orbit, linear-loop reachability, algebraic target hitting, and ultimate positivity;
10. a high-rank vector family showing genuine advantage over scalar methods.

This is not a breakthrough on the general Skolem Problem. It is a rigorous, checkable, and partially automatic framework for proving decidability in high-rank, high-order certified families.

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